

PART I



Understanding High-Intensity Interval Training

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Genesis and Evolution of High-Intensity Interval Training



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from Jean Claude Vollmer

Of all the factors related to sport performance that you'll ever read about, coaches and sport professionals will unequivocally agree that it is the preparatory training completed that has the greatest impact on sport performance. The preparatory training includes the highly important mental aspect, but additionally skill or technique development, tactical intelligence, and, of course, physical development. With training, we're preparing the body for war on the battlefield: the pitch, the court, the field, the road, the track, or in the water. So for athletes, coaches, and practitioners passionate about maximizing performance, investing thought and time toward planning in this area is worthwhile.

To prepare physically, varying amounts of different types of training will be important, depending on the context of the sport or event that is attempting to be optimized for, and this important component of the performance enhancement puzzle will be expanded on shortly. For now, we can say generally that varying appropriate amounts of both high-intensity training as well as continuous low-intensity training are critical ingredients needed to maximize performance in nearly all athletes who need a blend of prolonged endurance and high-intensity speed or power (16, 24). While there is little doubt that varying amounts of both types of training can effectively improve different aspects of performance-related

physiological function and physical development, the focus of this book of course will be on high-intensity interval training (HIIT) and how it can help to solve important aspects of an athlete's complex training program puzzle. When we consider the relatively simple adjustments to training programs that focus on submaximal aerobic endurance design, it's easy to see that the modification task involved (i.e., progressively increasing exercise duration) pales in comparison to that of HIIT format selection, where multiple moving parts within the format (intensity, duration, recovery, etc.) create immense complexity in design possibility and related physiological outcomes.

Defining HIIT

Let's begin with a simple definition. HIIT is usually defined as exercise consisting of repeated bouts of high-intensity work performed above the lactate threshold (a perceived effort of "hard" or greater) or critical speed/power, interspersed by periods of low-intensity exercise or complete rest. As we will outline, there are many ways that definition can be achieved, however, figure 1.1 provides a broad illustration of the concept. The general basis of high-intensity interval training can be described simply as follows: Imagine performing a bout of exercise at an intensity

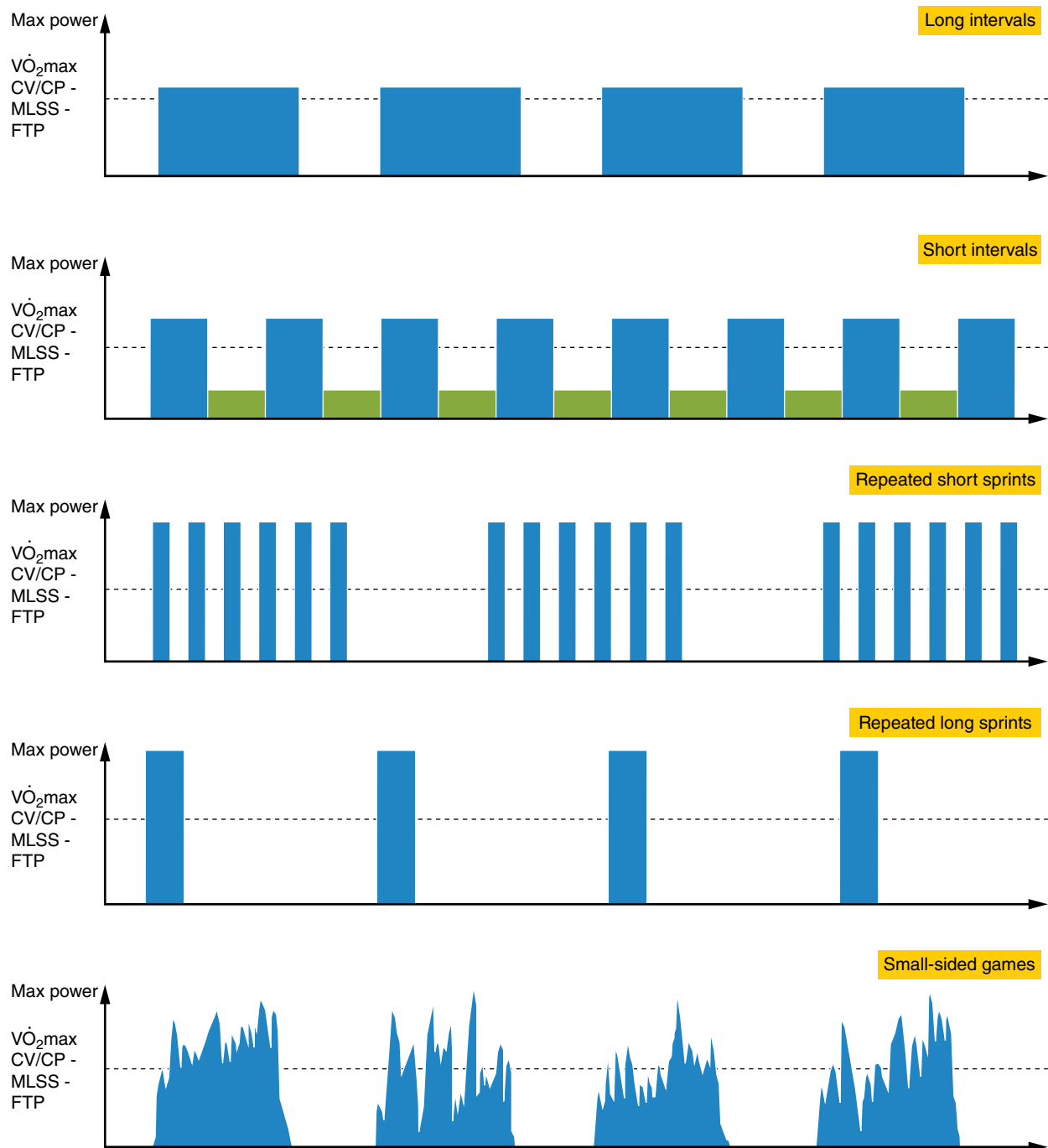


Figure 1.1 Schematic diagram illustrating the general concept of high-intensity interval training (HIIT) defined as repeated bouts of high-intensity exercise performed above the lactate threshold or critical speed/power, interspersed by periods of low-intensity exercise or complete rest. The five main HIIT formats, including long intervals, short intervals, repeated sprints of short and long and game-based training durations, are shown. Blue bars = effort intervals, green = relief intervals.

above your lactate threshold, or critical velocity/power (refer to chapter 4). To be clear here, this is an exercise intensity that is unsustainable, and one at which your brain would eventually force you to lower your intensity if you were to sustain it for as long as you could. It feels hard, and you know that fatigue would be inevitable at this pace if you were to hold on. Higher levels of sugar-burning glycolysis are needed generally to sustain the energy demand, and lactate accumulates to high levels typically at the point of fatigue. Now, if we took that same high-intensity effort and separated it with pauses that included periods of complete rest or lower levels of active recovery, that glycolytic energy rate is eased so that lactate production is more in check, whilst the cardiovascular strain remains high, and perceived effort, although still high, is reduced and manageable. This is one of the features of HIIT that absolutely fascinated us when we first started research into the topic. As shown early by Per-Olof Åstrand and colleagues who worked with him (2, 10), metabolic rate can be as high as the maximal rate of oxygen uptake ($\dot{V}O_{2\max}$), yet reasonably tolerated at relatively low concentrations of blood lactate (i.e., $<4\text{mM}$). From this work and others in the field, interval training was originally proposed as a method that allowed athletes to do greater volumes of conditioning work with less physiological strain. Ultimately, this broken exercise format of HIIT enables either a more manageable training session with the same training stress, or alternatively, the accumulation of a greater total amount of a high-intensity exercise stimulus if the sequence continues to be repeated. To be clear, by performing such high-intensity work intermittently instead of continuously, a person can maintain a high-intensity stimulus for longer, with less accumulated physical strain, and with beneficial adaptations that can be specific to sport demands. While interval training can be associated with a high degree of physical effort, fatigue, and acute discomfort, when applied properly with adequate recovery, it clearly has been shown to elicit rapid improvements in various aspects of performance and physiology.

History of HIIT

With Jean Claude Vollmer

Interval training has a relatively long tradition. Rumor has it that it may have started sometime around the year 1912. Based on advice received in letters from his brother, another professional runner

living in the United States, Hannes Kolehmainen of Finland, was seen preparing for races using 5 to 10 repetitions at his specific 10 km race pace of 3:05/km (19 km/h), before winning the 10,000 m event at the Olympics. In the 1920s and 1930s, coach Lauri Pihkala continued the development of these earlier HIIT forms with Paavo Nurmi (figure 1.2), one of the best middle- and long-distance runners in the world at that time. Nurmi was known as the “Flying Finn” and set 22 official world records at distances between 1500 m and 20 km, winning nine gold and three silver medals in his 12 Olympic Games events. His daily routines (26) were as follows:

- Early morning: 10 to 12 km walk interspersed with a few sprints, followed by generic gym exercises and a bath.
- One hour later, main training as: 4 to 5 $\times$ 80 to 120 m sprints, then 400 to 1000 m high-speed running, and finally 3000 to 4000 m high-speed running.
- Evening training: 4000 to 7000 m cross-country run, followed by 4 to 5 $\times$ 120 m sprints.

In the 1930s, Swedish coach Gösta Holmé is said to have invented *Fartlek*, meaning speed play



Figure 1.2 Paavo Nurmi. Bettmen/Getty Images

running, in Swedish. Such training is documented as being used often, from then into the early 1940s, to prepare a number of Swedish endurance runners and cross-country skiers, including modifying this type of training across various terrain types, termed *natural interval training*.

An example of a Fartlek session at that time might include the following:

- 1. Easy run for 5 to 10 min
- 2. 1200 to 2400 m fast run at a regular tempo
- 3. 5 min fast walk
- 4. Repeated 50 to 60 m sprints (separated by a recovery jog until the athlete feels ready to start again with the same quality). As many as needed to “feel” fatigue, before a 5 min walk
- 5. Acceleration starts over 3 to 4 strides, as explosive as possible
- 6. 160 to 180 m uphill running
- 7. Recovery jog

Athletes were allowed to repeat these seven points as they wished, but were instructed to ensure that they always felt fresh. When perceived fatigue approached a high level, they were requested to stop the session, which could be performed up to five times a week.

Following these earlier developments in Finland and Sweden, the German coach Woldemar Gerschler continued to popularize the use of intervals with the runner Rudolf Harbig, who established an astonishing 800 m world record in 1939 (1:46.6). Some sample sessions from that runner are shown in table 1.1.

HIIT became popularized further in the 1950s after triple gold Olympic (1952 Helsinki) champion in the 5000 m, 10,000 m, and marathon events, Emil Zátopek from Czechoslovakia (figure 1.3), described his training. According to Ettema (13), from early 1943, with his coach Josef Hron, Zátopek performed short interval training, running at paces near his critical velocity, calculated from his personal best

Table 1.1 Sample Session Prescribed by German Coach Woldemar Gerschler to Runner Rudolf Harbig to Establish His 800 m World Record in 1939

Date	Session
April 13	Jog 30 min 4×200 m in 23.8, 23.8, 24.3, 24.0 s (recovery 5 min walk) Jog 10 min 600 m in 1:25.3
April 23	Jog 30 min 800 m in 2:11 Jog 15 min 800 m in 2:02 Jog 10 min 600 m in 1:28
May 2	Jog 20 min 5×200 m in 23.2, 23.8, 24.6, 24.7, 24.2 s (recovery 5 min walk)
June 1	Jog 20 min Crouch sprint starts: 2×30 m, 2×50 m, 2×80 m, 1×150 m, 1×200 m, 1×400 m
June 7	Jog 25 min 2×300 m in 38.2 s (recovery 5 min jog) Jog 10 min 500 m in 1:11 Jog 15 min 200 m in 24.0 s
July 13*	Jog 20 min 600 m in 1:27 Jog 10 min 300 m in 36.9 s Jog 10 min 500 m in 1:06.7

* Last session before 800 m world record on July 15.



Figure 1.3 Emil Zátopek. Bettmen/Getty Images

in 3 to 10 km events. This was estimated to be 85% of his velocity at $\dot{V}O_2\text{max}$ ($v\dot{V}O_2\text{max}$; 20 km/h, or 1:12 per 400m), with daily training that included $5 \times 200 \text{ m} + 20 \times 400 \text{ m} + 2 \times 200 \text{ m}$ (for a total of 10 km), or the famous $100 \times 400 \text{ m}$ interspersed by 200 m recovery runs (45 s to 1:30). Remarkably, it was reported that Zátopek was already playing with some of the HIIT variables we will expand on in chapter 4, to modify the physiological responses during his winter build-up phase. That is, he would lower the intensity of the exercise runs and increase the intensity of his recovery jog (i.e., decreasing the amplitude), which would increase and decrease reliance on aerobic and anaerobic systems, respectively. At this point in time, Zátopek and Hron were already shaping their HIIT sessions to solve their performance puzzle.

From 1952 to 1956 in Hungary, coach Mihály Igloi and his runners Sándor Iharos (one of only two athletes with Paavo Nurmi to have held outdoor world records over 1500 m, 5000 m, and 10,000 m), István Rózsavölgyi (world record holder over 1000 m, 1500 m, and 2000 m in 1956), and László Tabori (on September 6, 1956, he equaled the 1500 m world record—set only a month earlier by

Iharos—with a time of 3:40.8), pushed world records further using forms of HIIT repeated twice a day, extending the volume of repetitions run at high-intensity. After the 1956 Melbourne Olympics, Igloi immigrated to the United States and spread further his methods at the Los Angeles Track Club. Among other runners, he coached Bob Schul, who won Olympic Gold in Tokyo (1964) in the 5000 m. An example of a weekly microcycle for Rózsavölgyi in April is shown in table 1.2.

Around the same time, Franz Stampfl, an Austrian who left the German Nazi regime, helped promote repetition work in England with his followers Chris Chataway (5000 m record holder in 1954; 13:51.6) and Chris Brasher (Olympic Gold Medalist in the 3,000 m steeplechase in 1956 in Melbourne; 8:41.2) and Roger Bannister; the former men paced Bannister to break the 4 min mile record in 1954.

While Kolehmainen, Nurmi, Harbig, Rózsavölgyi, and Stampfl had reportedly used sets of high-speed repetition runs (called “repetition work”), none, with the exception of Zátopek, were yet training using actual HIIT sequences that incorporate much shorter recovery periods between repetitions and larger series volumes. It was only later into the 1950s when physiologists made suggestions to coaches and their athletes to shorten recovery intervals. These recommendations were based on heart radiographies and ECG data from hundreds of athletes. Their combined goal was to increase the level of acute physiological response during training.

At that time in fact, the main physiological benefits of HIIT were thought to arise during processes occurring within the recovery periods, a phenomenon we will highlight in chapters 3 and 4. According to Billat (4), interval training was first described, in a scientific journal, by physiologists Reindell and Roskamm (23), who were inspired through discussions with coach Woldemar Gerschler, and together created the Fribourg interval training method. Different HIIT formats (that in our book we will begin to term *weapons*), were presented at various coaching conferences (1), which tended to be either long (e.g., $6\text{--}10 \times 600 \text{ m}$ in 1:38–1:42 with recovery periods of 1:30–2:00, or $6\text{--}8 \times 1200 \text{ m}$ in 3:24–3:30 with recoveries of 1:30–2:00) or short intervals (e.g., up to 100 reps of 100–200 m runs in 14–16 s and 30–33 s interspersed with 30 to 60 s rest periods). In 1960, the pioneering Swedish physiologist Per-Olof Åstrand described the cardiorespiratory responses to long interval training (3 min efforts) at a velocity

Table 1.2 Sample Weekly Microcycle Used by István Rózsavölgyi in His Lead-Up to the 1956 Olympics

Day	Training
1	2×4×300 m in 45.0 s (recovery 100 m jog) 10×100 m in 15.0 to 18.0 s (recovery 50 m jog) 6×300 m in 45.0 s (recovery 5 min jog) 10×100 m alternatively in either 16.0 to 18.0 s or 14.0 to 15.0 s (recovery 50 m jog)
2	5×5×200 m in 27.0 to 28.0 s (recovery 100 m jog) 6×100 m in 15.0 s (recovery 50 m jog)
3	10 to 15 km fartlek 15×100 m (recovery 50 m jog)
4	5×300 m easy, 5×300 m fast, 5×300 m very fast, 5×300 m easy 10×100 m in 16.0 s (recovery 50 m jog) 6×300 m at 80% personal best 10×100 m fast, 10×100 m easy
5	10×10×100 m (alternatively high and moderate speed between series)
6	10×150 m moderate pace 2×3×400 m in 55.0 to 56.0 s (recovery 300 m jog between reps and 400 m walk between series) 10×100 m (recovery 50 m jog) 10×100 m high tempo (recovery 50 m jog between reps and 400 m walk between series)
7	Cross-country race

of 90% to 95% $\dot{V}O_{2\max}$, known already to elicit $\dot{V}O_{2\max}$ in the last repetitions, despite complete passive rest intervals. Åstrand et al. (2) considered that this was one of the best forms of interval training to improve $\dot{V}O_{2\max}$, since all cardiorespiratory parameters were at their maximum. Remarkably, Christensen and colleagues (10) showed that very short interval training, consisting of 10 s runs at 100% of $\dot{V}O_{2\max}$ interspersed with 10 s of complete rest, could elicit $\dot{V}O_{2\max}$ under low blood lactate levels (<4 mmol).

In the late 1960s, the American group of Fox, Robinson, and Wiegman (14) were one of the first to compare metabolic energy use during continuous and interval running at the same exercise rate, and like others, showed a slower accumulation of lactic acid, and delayed fatigue with interval training, speculated to be due likely to the replenishment and subsequent reutilization of phosphagen reserves via the muscle's oxygen-storing myoglobin. As a result, athletes were shown to be able to accomplish larger quantities of work at very high intensities.

In the field, the 1960s saw a large development of HIIT methodology and especially its use within block periodization approaches, thanks to, among others, Australian Percy Wells Cerutti, coach of Herb Elliott (1500 m Olympic Gold in Rome 1960 in 3:35.6) and New Zealander Sir Arthur Lydiard (coach of Peter Snell, triple Olympic Gold medalist, and only male since 1920 to win both the 800 m (world

record in 1962 on a grass track, 1:44.3) and 1500 m at the same Olympic Games, in Tokyo 1964 (18).

The 1970s and 1980s saw more athletes using interval training in the field. More thought and reflection appeared to spread, as the preparation of training began to be considered more and more like a puzzle that needed solving. Different forms of training were trialed and tested, including long slow running of great distance (24), as well as various specific paces thought to enhance physiology or prepare optimally for competition. These philosophies, inspired largely by Lydiard (18), evolved today into what is more commonly referred to as “polarized training” (24). Here, roughly 80% of an athlete's training volume is performed at an intensity below his so-called aerobic threshold (elaborated on in chapter 3), while the remaining 20% of the training, including HIIT, is performed above this intensity. The integration of HIIT into an athlete's complete training system gathered momentum and acceptance throughout the world, with Franck Horwill implementing a five-speed system with Sebastian Coe, who won four Olympic medals, including 1500 m gold at the 1980 and 1984 Olympic Games. Paul Schmidt in Germany and Moniz Pereira from Portugal continued to develop the idea in the 1980s that HIIT was a necessary part of a runner's training program puzzle. Commercialization and professionalization within running led to further improvements in understanding HIIT responses and the effect of

variable manipulation, with anecdotal reports from coaches on the different effects of manipulating training intensity, duration, repetitions, series, inter-reps and inter-series recovery modalities, as well as terrain (hill, grass, sand). It was not until the 1990s that more detailed work on interval training was published for the masses by Tim Noakes, in his famous book *The Lore of Running* (21) and Jack Daniels with *Jack Daniels' Formula* (12). Subsequently, most other individual athlete sports naturally followed the lead of running and adopted similar training tactics, including cycling, swimming, rowing, cross-country skiing, and triathlon (see related chapters).

The year 2000 began an explosion of work in the area of interval training research, beginning with Billat's work that focused on aerobic (4) and anaerobic (5) interval training for runners, and Laursen and Jenkins (17) reviewed the physiological basis for interval training across individual athlete-based endurance sports. An outpouring of academic research followed (literally thousands of papers on HIIT) from both areas of sport application (6, 7) and exercise science (19) thereafter.

Team Sports

While research performed across traditional individual Olympic-based sports shows direct correlations between various levels of physical capacity and performance, the relationship for team and racket-based sports outcomes (i.e., winning a match or scoring a goal) is not as straightforward. Nevertheless, a rigorous preparation of physical capacities has always been believed to be of high importance for successful outcomes. Historically, team sport coaches often had backgrounds as physical education teachers and were therefore well versed in the physical preparation principles that they borrowed from track-and-field coaches. For this reason, HIIT has nearly always been incorporated into team training schedules. Examples include run-based formats directly taken from track-and-field practices (series of 100 and 200 m efforts, up to 1 km repetitions, or using 15 s on/off formats), or even using more sport-specific approaches, namely game-based HIIT (figure 1.1), with athletes practicing their sports at high-metabolic work intensities alongside specific rules and typical sport constraints, often termed small-sided games in the team sport context (22).

In the 1970s in the United Kingdom, Tom Reilly, who some have termed “the father of football

(soccer) science,” alongside his pioneering work in monitoring, training-load quantification, and player education, showed additionally that HIIT may be important for some aspects of team sport performance (22). Around the same time in Italy, a strong athletic culture led by Enrico Arcelli used modifications of the track-and-field HIIT approach in their practice. This work, which used HIIT, paved the way for others within the Italian Association of Fitness coaches. Also in Italy, the famous Milanello training center, the Milan Lab, was the first institution fully devoted to sports science in football, embracing innovation and technology. Following Reilly, the Scandinavian group of Bjorn Eklom, Jens Bangsbo, and Paul Balsom published much work in the area, summarized in Bangsbo's 1994 book, *Fitness Training in Soccer: A Scientific Approach* (3), which led the basis of interval training for soccer. Subsequently, most other team sport codes followed the lead of football and adopted similar training tactics.

The progressive development of team sports into the mass entertainment entities they now are came with an explosion of staff employed to maximize performance and outcomes (see, for example, Part II). Team sports, and more especially top football (soccer) clubs (e.g., Manchester United, Arsenal, Liverpool, and Paris SG), youth academies (e.g., Aspire in Qatar), as well as other football codes including AFL and Rugby and, more recently, various U.S.-based sports (basketball, American football, baseball, and ice hockey), today employ physical conditioning specialists alongside sport science research specialists, including physiologists, biomechanists, statisticians, and technology support experts. Similar teams of experts are amassed in today's leading country institutes and academies focused on individual athlete performance (cycling, track and field, swimming, rowing, triathlon, cross-country skiing, etc.), preparing for Olympic games and world championship performances. These physical preparation teams' combined roles are to establish noninvasive monitoring systems and advise on areas of potential performance enhancement. As a result of both our increased knowledge and the explosion of human resources, HIIT practices at the elite level may now be less empirical, but pursuant of specificity and individualization, replaced by monitored and tailored sequences across the most relevant HIIT formats (8, 9, 15).

In 2013, we published a two-part review (6, 7) that combined our applied work as embedded practitioners in high-performance sport, along with our

academic research experience. It was an expansion of the ideas outlined in these publications alongside the assistance of Dr. Jackson Fyfe to navigate us through the complexities of the strength training interference phenomenon (chapter 6), Dr. Phillip Maffetone to ensure we understand HIIT's effect from a holistic standpoint (chapter 7), in addition to outlining methods we can use to program the load of HIIT (chapter 8) and monitor its response (chapter 9), that form part I of this book. The second half of our book is dedicated to the application of these principles for, give or take, the world's top 20 sports that typically apply HIIT in their practice. Each application chapter in part II, generously contributed by some of the best practitioners and coaches in the field today, additionally provide tools you can use to maximize athletic performance in these sports. Combined, both science and application sections expose opportunities for future investigation by today's students and researchers in academic institutions throughout the world. We hope that you, the reader, enjoy its contents and find it useful.

Performance Enhancement Puzzle

While the focus of our book is on high-intensity interval training, we're mostly all here to learn how we can use HIIT to improve the physiology that maximizes performance. From a physical preparation standpoint, what can be gained with HIIT that could enhance performance? Perhaps before we go there, we need to pause for a moment to gain clarity around an important question. That question is: What is performance? Performance probably means something a little different for everyone. For MB, in his home environment, performance might mean that his team is able to recover quicker between high-intensity actions while maintaining a higher playing tempo and better tactical alertness during a football match, alongside more goals scored over his opposition. For PL, performance might mean he's seeing an already high cycling power output and running speed enhanced in training alongside improved finish time in his triathletes. You'll have your own point of view depending on where you're coming from, and what's meaningful to you, as will all the sport-specific contributing experts in the application chapters that accompany the second half of our book. Typically, however, if we want to put a general definition on performance in the context of most sports, we refer to the ability to carry out or accom-

plish an action, task, or function. To do so, we must usually keep fatigue at bay. In this regard, HIIT is an important component of most sports' training programs, as it's a training format that allows specific preparation toward meeting the physical demands of performance. Thus, when we're referring to how HIIT fits into the programming puzzle, we mean that we're trying our best to select the HIIT sequence that best enhances physiological adaptation or outcome. These adaptations or outcomes from an HIIT session should be preparing athletes or teams optimally, placing them in the best possible physiological state they can be in before they perform in their event or match.

Before moving forward with HIIT design and shaping the puzzle piece, we need to take a step back first and ask a few more questions. First of all: How important is physical preparation in the big picture of our sport? We need to ask this question, as it provides us clarity as to where we should be focusing our time and energy. If you're reading this book, then it's likely that physical preparation is your specialty and of high interest. Regardless, it's important to appreciate that physical preparation might not actually be the critical thing that is necessary to maximize your sport's outcome (hard to believe, we know). In the context of some sports, in order to be a valued member of your team, it's important to have clarity in this regard, so that appropriate focus can be made toward the things that matter. The important first question then is: How important is physical preparation relevant to other critical factors that relate to sporting success, such as skills and tactics? Figure 1.4 provides an example of how this differs throughout the sports presented in our book.

Once we are clear on the level of importance of physical preparation, it's now time to begin exploring where HIIT fits into the physical preparation puzzle matrix. Here again, HIIT is a critical weapon we can use to shape the puzzle piece, but it can't be considered the only form of preparation. HIIT is one of a number of different components of the programming puzzle that we must consider, with different emphases varying dependent on the nuances of the sport. Skill development, technique, match intelligence, tactics, coordination, pacing, agility, recovery, strength, explosive power, endurance, repeated sprint ability, anaerobic speed reserve, tempo training, and others, must also be considered alongside HIIT.

But sport requires energy and various forms of fatigue resistance. Having these features in an athlete is critical in so many sports and is the hallmark of

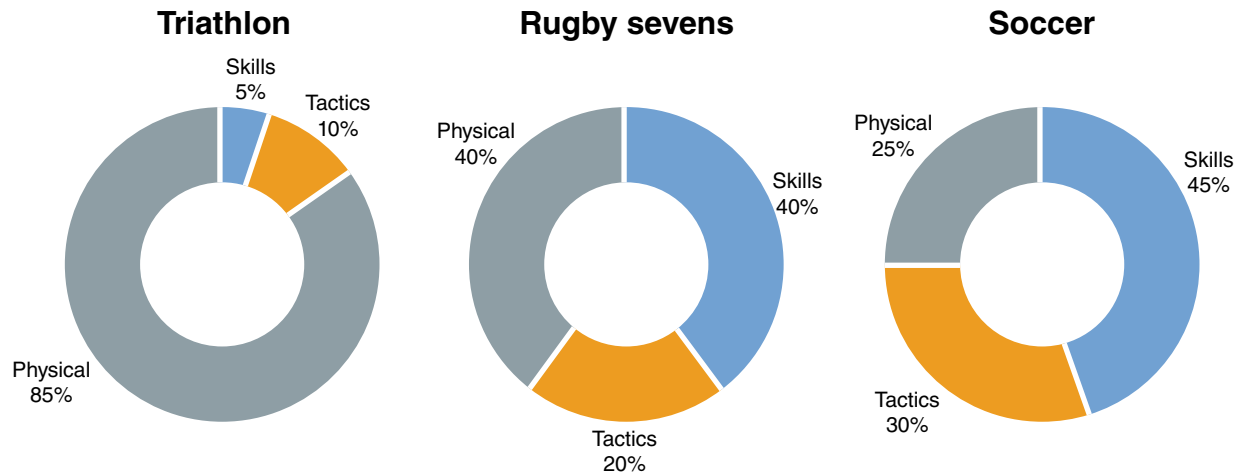


Figure 1.4 Respective importance of skills, tactics, and physical capacities for successful performance in three competitive sports. Overall estimation for all field positions pooled.

successful athletes. Thus, HIIT is often the critical tool or weapon we can use to shape the puzzle piece to achieve optimal physiological state and subsequent performance probability for a number of sports that are of interest on the world stage. In this book, our aim is to outline the science that explains how the weapons work and how they can be manipulated to hit our physiological targets. Our expert sport practitioners will guide us toward their use of HIIT at the coalface of elite sport in our application chapters.

HIIT is highly complicated, and to solve the puzzle, you need to shape the puzzle piece appropriately to solve the fit. Our aim is to provide you, the reader, with the tools to shape this important piece of your programming puzzle.

There are many ways that we can shape an HIIT puzzle piece (see chapter 4), but only so many options we have to solve the fit appropriately. Consider the analogy of baking a cake. For any cake, the ingredients are pretty simple and largely the same: flour, butter, sugar, eggs, and a bit of baking powder. But are all cakes the same? Clearly not. Preparing for a sport isn't that different in this context. Getting the raw ingredients together, like athletes, tactics, and skills, is relatively straightforward. But when we look to physical preparation, the number of moving parts can make programming rather daunting. Thus, the challenging aspect of baking a cake for many sports is how best to prepare athletes physically, using intelligent programming and rapid adjustment in line with forever-changing circumstances.

Switching analogies, and back to the programming puzzle. Where does one begin? Figure 1.5 illustrates the main considerations we need to make

before designing our HIIT sessions. Once we're clear of the importance of physical preparation for our sport (figure 1.4), we next need to look at:

1. the sport's physical demands,
2. athlete profile(s),
3. long-term adaptation targets, and
4. training periodization or how the training plan fluctuates daily, weekly, and season long, depending on our aims and the sport/event schedule.

All these factors merge to establish the desired physiological objectives of the HIIT session. Once we're clear on our physiological aims, which establish which of the five physiological targets we're aiming to hit (HIIT types), we can then appropriately choose from five broad HIIT formats (weapons) and their manipulations (figure 1.5), which we will expand on throughout part I of the book.

Consideration 1: Sport Demands

First and foremost, we need to understand the required systems to enhance relevant to the sport in question. What are the sport demands? How does meeting these demands enhance probability toward hitting our performance target?

Do we need a good level of aerobic power/endurance due to the duration of the activity? Does the sport involve high-intensity supramaximal anaerobic efforts? High-level strength? Speed? Repeated intermittent high-intensity efforts? Combinations of many of these features? Every sport is different, but we first

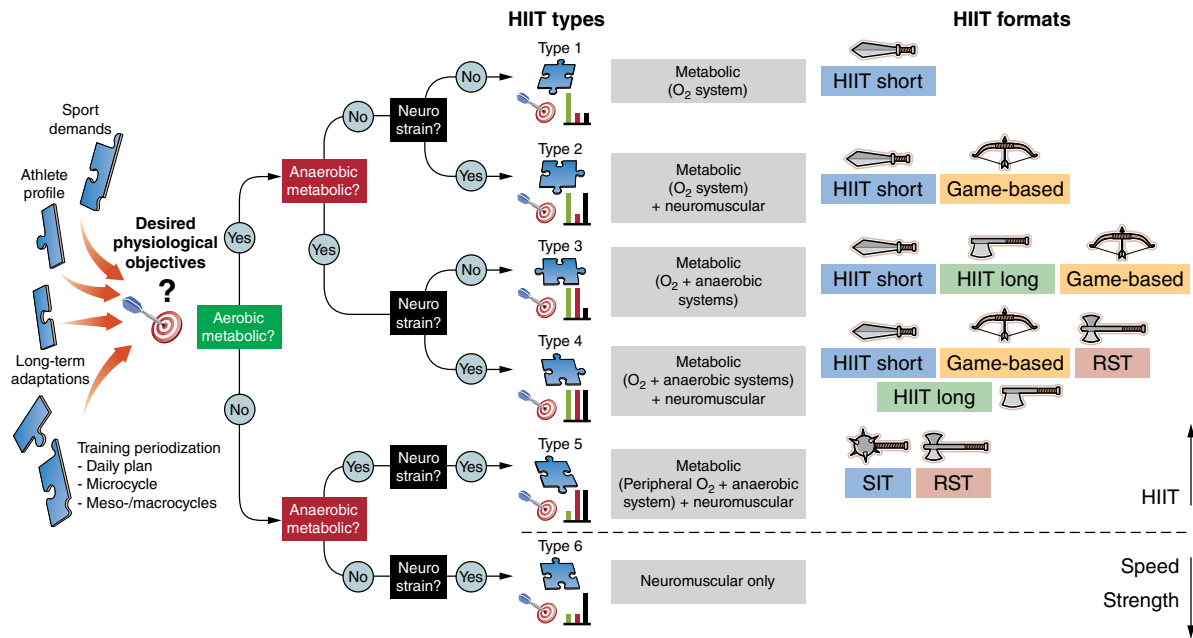


Figure 1.5 Factors to consider to appropriately choose an HIIT session type include the sport demands, the athletes the training is being applied to, the desired long-term adaptations, as well as the periodization aspects. Together, these determine the desired physiological response (aerobic metabolic, anaerobic metabolic, neuromuscular) to target. These six physiological response targets are divided into the following categories (used throughout): type 1, aerobic metabolic, with large demands placed on the oxygen (O_2) transport and utilization systems (cardiopulmonary system and oxidative muscle fibers); type 2, metabolic as type 1 but with a greater degree of neuromuscular strain; type 3, metabolic as type 1 with a large anaerobic glycolytic energy contribution but limited neuromuscular strain; type 4, metabolic as type 3 but a high neuromuscular strain; type 5, a session with limited aerobic response but with a large anaerobic glycolytic energy contribution and high neuromuscular strain; and type 6 (not considered HIIT) involving a high neuromuscular strain only, referring typically to speed and strength training. The HIIT formats (weapons) that can be selected to appropriately hit the HIIT target-type responses include short intervals, long intervals, repeated sprint training, sprint interval training, and game-based HIIT, described in detail in chapters 4 and 5.

Adapted from M. Buchheit and P.B. Laursen, “High-Intensity Interval Training, Solutions to the Programming Puzzle: Part I: Cardiopulmonary Emphasis,” *Sports Medicine* 43, no. 5 (2013): 313-338.

must gain clarity around what it is we’re trying to create in our athletes before we design the program. As one example, consider that an 800 m runner would likely favor a greater proportion of “anaerobic-based” HIIT compared to a marathon runner.

Making observations relative to how the ideal athlete looks from a physical standpoint, in addition to reading literature in the area, may be a prudent starting point.

Figure 1.6 shows the key sports described in our book and how the importance of the three main physical capacities (speed, strength, and endurance) may vary with respect to each other from sport to sport. Thus, HIIT type, volume, and programming should be made appropriately in line with its impor-

tance. HIIT in its various formats (figure 1.5), provides weapons we can use to hit the physiological targets of importance for each athlete and their sport (figure 1.6).

Consideration 2: Athlete Profiles

We humans are all unique. We all have our strengths and weaknesses. This makes us interesting. But this individuality creates an important consideration when we’re trying to build performance, either for single sport performers or especially when we have a team of athletes. Each athlete we come across may require a different HIIT strategy to affect her individual physiology targets. While all players may be of the same age, playing on the same

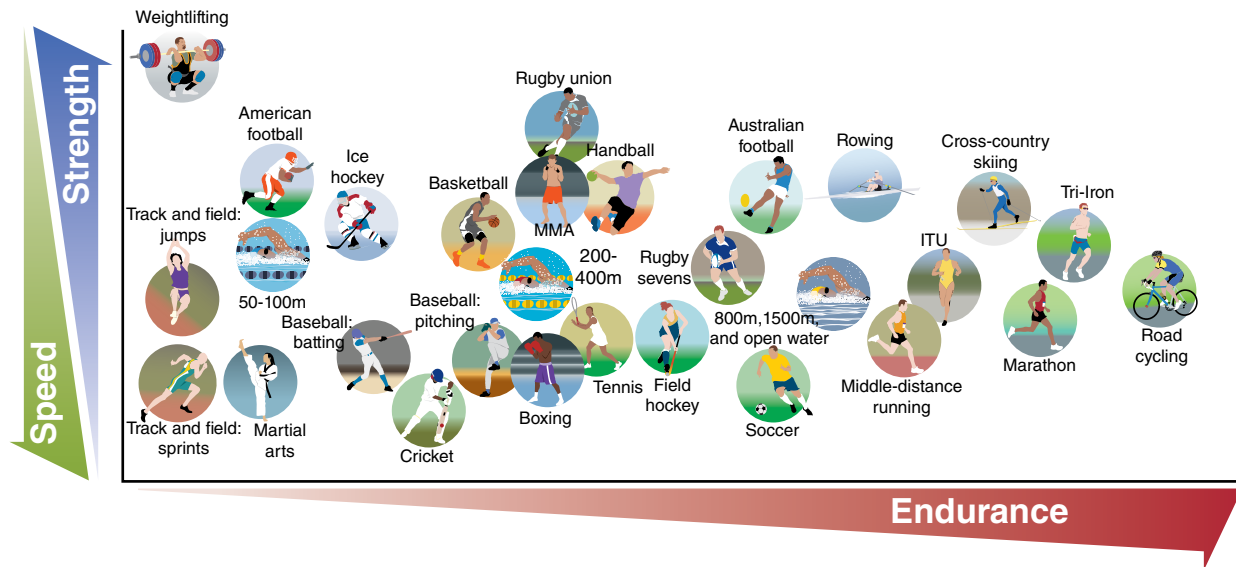


Figure 1.6 Various sports, their physical demands relative to needed speed, strength, and endurance.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

team, their physical capacities are notably dissimilar. As a result, the best HIIT type to select for each athlete would ideally be very different (figure 1.5). An added layer of complexity that practitioners must consider is that not only do differing athlete profiles create distinct individual needs but also that individual athletes encounter unique responses to the same HIIT format (11).

Considerations 3 and 4: Long-Term Adaptation Targets and Periodization

As mentioned, choosing the appropriate HIIT session to achieve the necessary physiological response for a given training cycle is no simple task, and some call this the art of good programming. This is ultimately what we hope you will learn most from this book. We want to teach the art and science of HIIT programming. Why should a short interval session go here, and repeated sprint training go there, and not the other way around? Where does strength training fit into the mix, and how does that impair or complement adjacent HIIT programming? What is the best way to shape the HIIT puzzle piece and its associated physiological response so that it appropriately fits into your weekly or monthly training block? Figure 1.7 provides an illustration of the concept from the world of elite football and road running using two different typical weekly patterns.

For both scenarios, the problems are clearly depicted. How do we best place HIIT across other important aspects of training throughout the week, such as endurance, strength, and speed sessions. How do we consider the impact of one session against the other, so that physiological outcomes of importance for our sport are maximized? Importantly, the physiological consequences from the appropriately selected HIIT session should provide the proper acute signaling response to elicit the optimal adaptations needed.

The general rules that guide HIIT selection for a given day/session include selecting:

1. the session's likely acute metabolic and neuromuscular response,
2. the associated cellular signaling that will target long-term adaptations,
3. the accumulated training load arising from other sessions that constrain/dictate the HIIT type,
4. the time needed to recover from the session (including the added HIIT sequence), and
5. its impact on the quality and ability to perform in subsequent sessions.

For these reasons, on a short-term basis, training periodization has a large impact on the HIIT prescription. Thus, many of the desired training adaptations are often considered as being training cycle

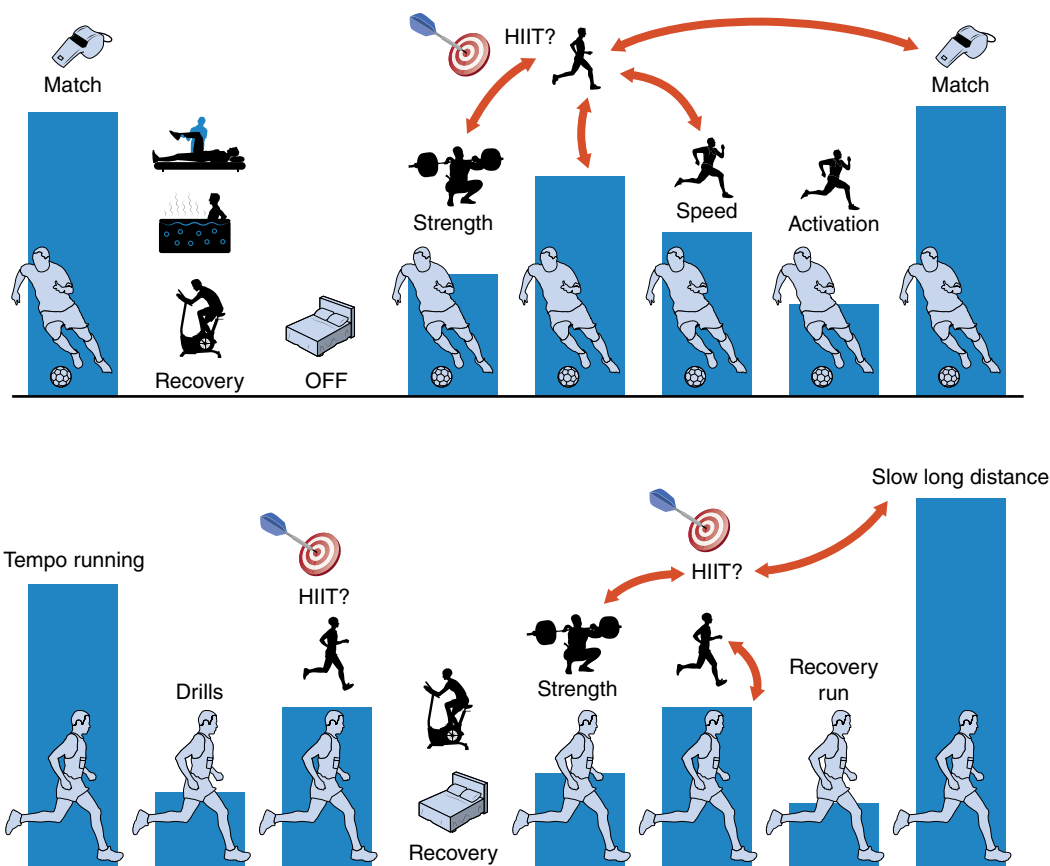


Figure 1.7 Two different typical weekly patterns (Team sports, with endurance, strength, and speed sessions on different days) and road running: day-to-day planning to shape the puzzle piece on a microcycle level, where the physiological objectives of the HIIT session must be considered in relation to the programming of other sessions. Vertical axis and blue shading illustrates the predicted training load (intensity $\times$ volume; see chapter 8).

dependent. To illustrate, within the team sport setting, generic aerobic power development might be emphasized in the initial phase of the preseason compared to more sport-specific and anaerobic-like HIIT sessions emphasized toward the start of the competitive season. Additionally, for athletes training twice a day, and/or in team sport players aiming to enhance metabolic and neuromuscular systems simultaneously, the physiological strain associated with a given HIIT session needs to be considered in relation to the demands of other physical and technical/tactical sessions so as to avoid overload and enable appropriate adaptation. Specifically here, our aim is to maximize a given training stimulus and minimize musculoskeletal injury risk. Thus, there are likely several approaches (i.e., HIIT formats) that, considered in isolation, will achieve a similar metabolic and/or neuromuscular training adaptation outcome. However, the ability of the coach and practitioner to

understand the isolated effects of various HIIT variables may assist with selection of the most appropriate HIIT session to apply to help solve the puzzle, at the right place and time.

Choosing the best solution to such a puzzle might be analogous to how an admiral goes about deciding which operation is best suited to take out a given military target, and we'll come back to this analogy throughout the text. While mass destruction weapons might hit all targets at once (e.g., training systematically very hard, "no pain, no gain approach," HIIT format type 4), collateral damage often occurs (analogous to added stress, extra fatigue, more injury/illness). Sometimes then, the best solution might involve specific "Navy SEAL-type" operations (John Rambo), for more specific targeting of the required physiological capacities, but with less risk of collateral damage. Along these lines, HIIT sessions should first be specific to the physiological adaptations desired

(figure 1.5) and not necessarily compulsory to the sport itself.

Summary

High-intensity interval training is usually defined as exercise consisting of repeated bouts of high-intensity work performed above the lactate threshold (a perceived effort of “hard” or greater) or critical speed/power, interspersed by periods of low-intensity exercise or complete rest.

The five main HIIT formats tend to involve long intervals, short intervals, as well as repeated sprints of short and long durations, as well as game-based HIIT.

To use HIIT appropriately, practitioners should:

1. Understand the context, including the sport, individual athlete needs and unique responses, phase of the training cycle, and the possible interference effects.
2. Define the target (desired physiological response/adaptation and, in turn, one of the six target types).
3. Choose (one of the) appropriate HIIT formats (HIIT short intervals, HIIT long intervals, game-based HIIT, RST, and SIT).
4. Manipulate the HIIT variables to shape the piece (described in chapters 4 and 5).
5. Monitor the load (chapter 8) and its response (chapter 9).

From its early beginnings on the track at the start of the 19th century, HIIT has reached exponential growth in popularity in academic research, high-performance sport, and the health-and-fitness sector, where today it is now considered the highest interest topic (25). Notwithstanding its high importance, HIIT is one of a number of tools in the coach's and practitioner's toolbox that they can use to enhance physical performance. Understanding the sport demands, individual athlete profile, and periodization phase is needed before we put pen to paper and prescribe the session. Here, we begin our journey into HIIT by describing its traditional methods of programming (chapter 2), the underlying physiological targets it affects (chapter 3), the factors we can manipulate to form HIIT weapons (chapter 4), and how to fine-tune them (chapter 5). But we are not finished there, as other factors of the training program puzzle that complement HIIT must be considered. To this end, Dr. Jackson Fyfe will navigate us through the complexities of strength training (chapter 6) and Dr. Phillip Maffetone will ensure we understand HIIT's effect from a holistic health standpoint (chapter 7) before we outline methods we can use to program HIIT load (chapter 8) and monitor its response (chapter 9). Once the science is outlined, we'll pass the baton to practitioners embedded in 20 of the world's most popular sports, and learn how they effectively apply these concepts in their practices at the coalface of the world's leading sports. We hope you enjoy the journey of understanding HIIT and its application as much as we have.

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